

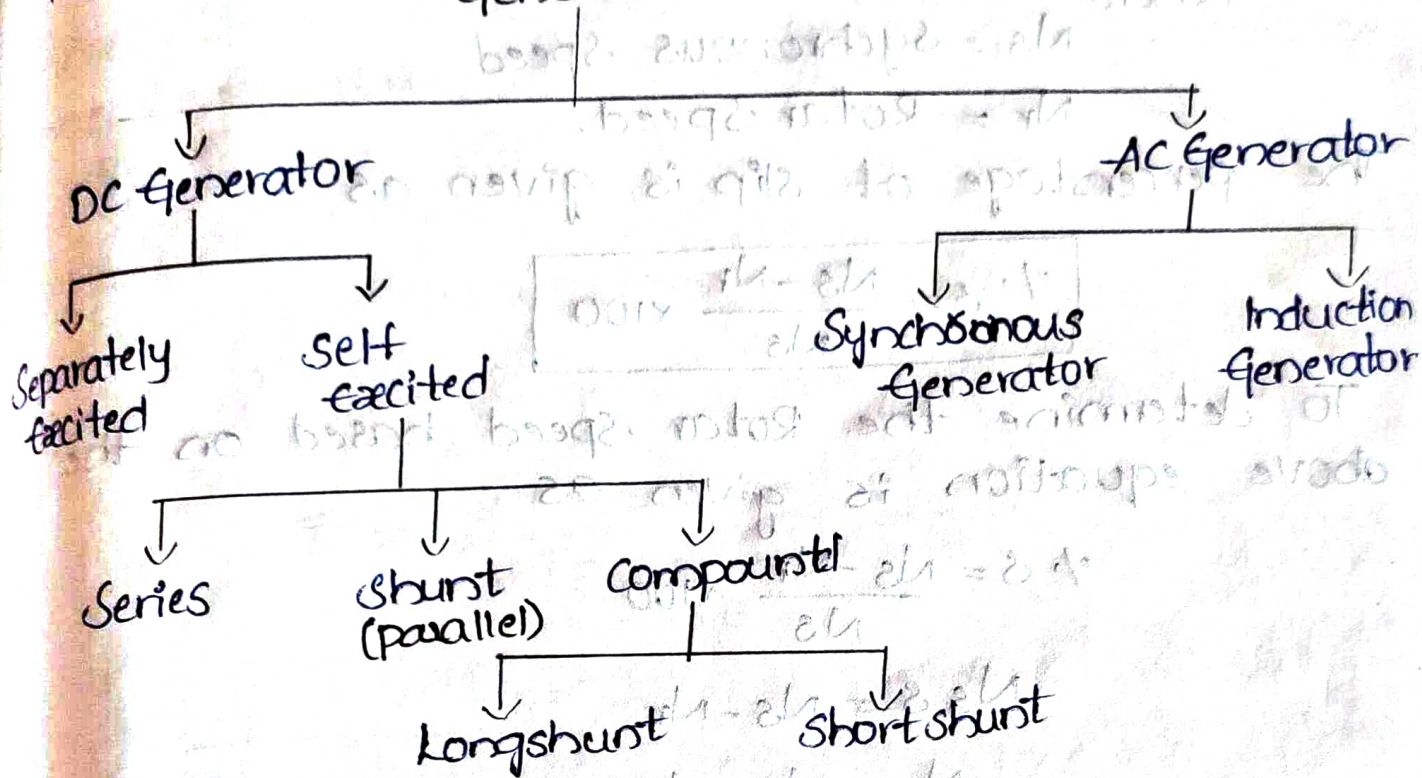
Unit - 4 Electrical Machines

Generator

Mechanical energy into Electrical energy is known as generator.

The classification of generator are

Generator

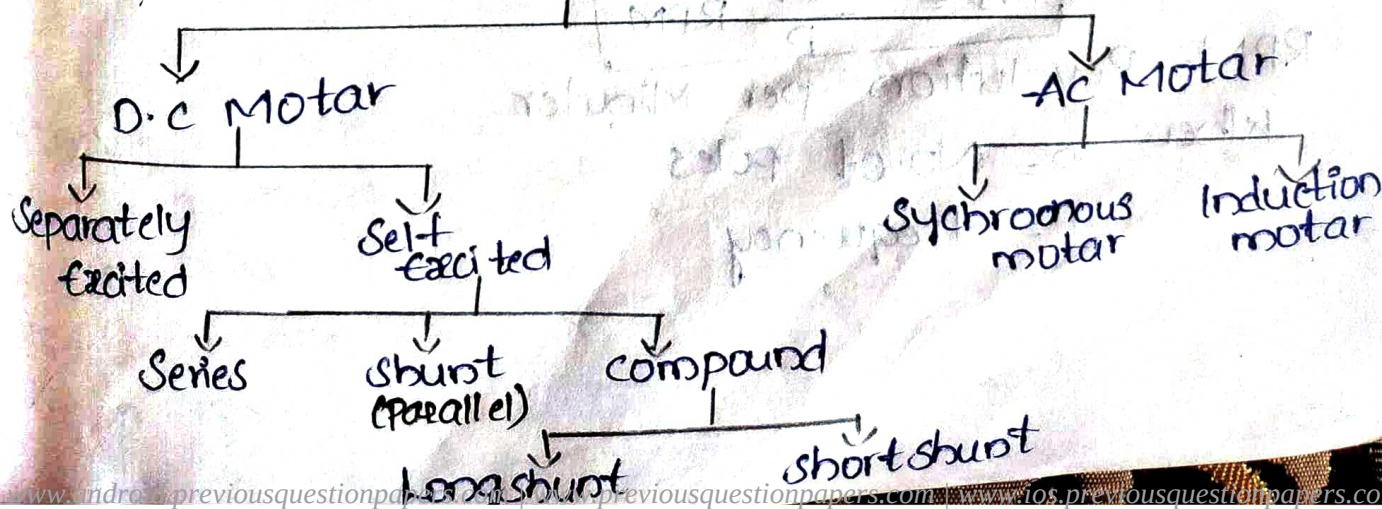


Motor:

In electrical machines which converts Electrical energy into Mechanical energy is known as Motor

The classification of motor are

Motor



Slip: It is defined as ratio of difference b/w Synchronous speed and rotor speed to Synchronous speed.

The mathematical expression slip is

$$S = \frac{N_s - N_r}{N_s}$$

Where

N_s = Synchronous Speed

N_r = Rotor Speed.

The percentage of slip is given as

$$\% S = \frac{N_s - N_r}{N_s} \times 100$$

To determine the Rotor speed based on the above equation is given as

$$\% S = \frac{N_s - N_r}{N_s} \times 100$$

$$N_s S = N_s - N_r$$

$$N_r = N_s - N_s S$$

$$N_r = N_s (1 - S) \text{ RPM}$$

Synchronous Speed:

It is defined as the speed which magnetic field rotates is known as synchronous speed. It is denoted by N_s

$$N_s = \frac{120f}{P} \text{ RPM}$$

RPM - Revolution per Minute

Where p = No. of poles

f = frequency

Rotar frequency

It is defined as the product of slip and actual frequency is known as Rotar frequency. It is denoted by f_r

$$f_r = s \times f \text{ Hz}$$

Q) Three phase induction motor has four poles is connected 400 V, 50 Hz supply and percentage of slip is 6% and calculate the Nr.

Given data

3- ϕ Induction motor

No. of poles = 4

Voltage $V = 400 \text{ V}$

frequency = 50 Hz

% Slip = 6%

$$= 0.06$$

Nr = ?

$$N_r = N_s (1 - s)$$

$$N_s = \frac{120 f}{p} \text{ rpm}$$

$$= \frac{120 \times 50}{4}$$

$$N_s = 1500 \text{ RPM}$$

$$N_r = N_s (1 - s)$$

$$N_r = 1500 (1 - 0.06)$$

$$N_r = 1410 \text{ rpm}$$

Q) Three phase induction motor 50 Hz and full load rotar speed 960 RPM calculate the

No. of poles

Slip frequency and consider synchronous speed

1000 rpm.

Given data

frequency = 50 Hz

Rotor Speed = 960

Synchronous Speed = 1000

$$N_s = \frac{120f}{P}$$

$$P = \frac{120f}{N_s}$$

$$P = \frac{120 \times 50}{1000}$$

$$P = 6$$

$$S = \frac{N_s - N_r}{N_s}$$

$$= \frac{1000 - 960}{1000}$$

$$S = 0.04$$

$$f_r = S \times f$$

$$= 0.04 \times 50$$

$$f_r = 2 \text{ Hz}$$

Q) 3- ϕ induction motor 400V, 50 Hz which 6 poles as rotor frequency 2 Hz calculate the slip and Rotor Speed.

Given data

Voltage V = 400

frequency = 50 Hz

Rotor frequency = 2 Hz

$$f_r = S \times f$$

$$S = \frac{f_r}{f} = \frac{2}{50} = 0.04$$

$$S = \frac{N_s - N_r}{N_s}$$

$$= \frac{1204}{P}$$

$$= \frac{120 \times 50}{6}$$

$$N_s = 1000$$

$$N_r = N_s (1 - S)$$

$$= 1000 (1 - 0.04)$$

$$N_r = 960 \text{ RPM}$$

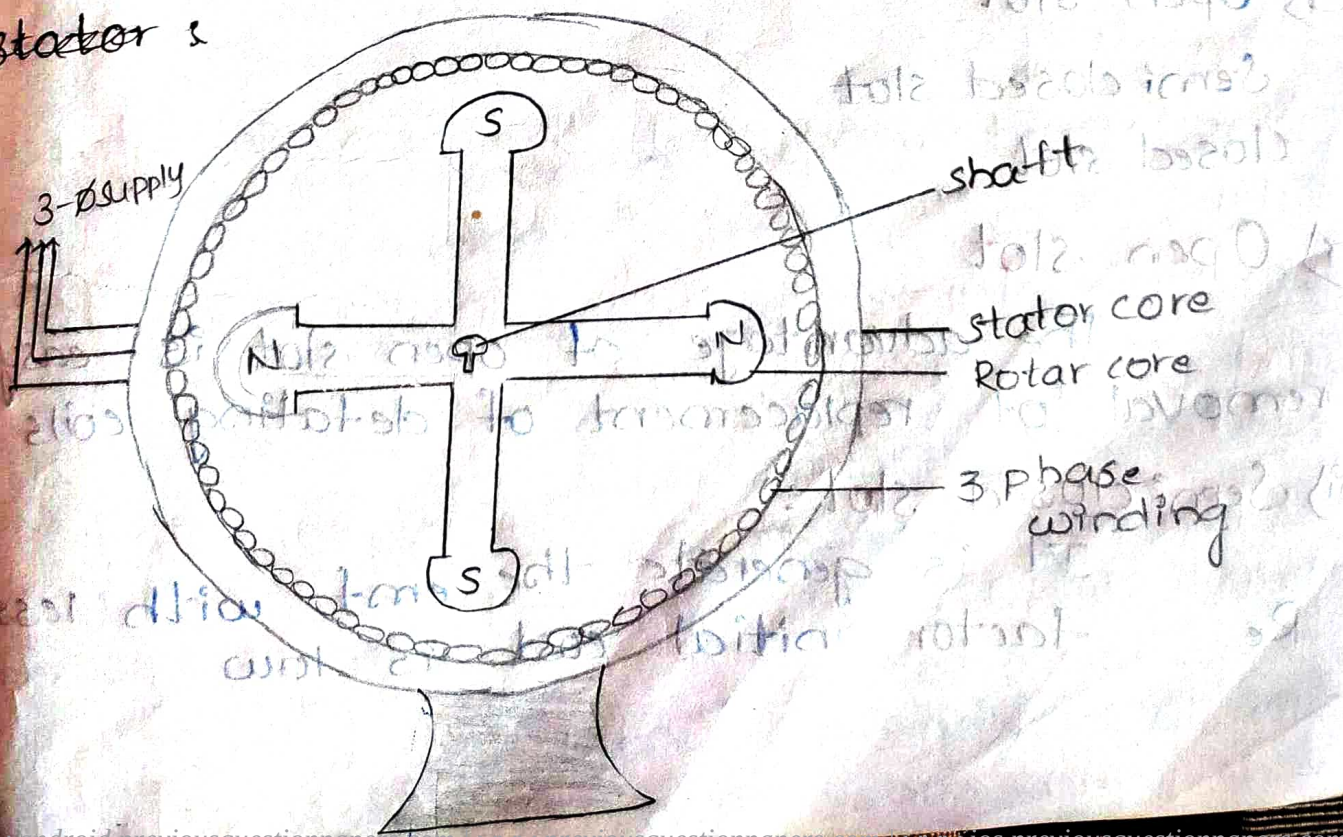
Construction of Synchronous Machines (or) Alternator
Synchronous machines is double excited energy conversion device. Since it is field winding is connected to dc supply and Armature winding deliver the Ac power.

Synchronous machines reduce the Iron losses and alternators consisting of two parts.

i) Stator [Armature winding]

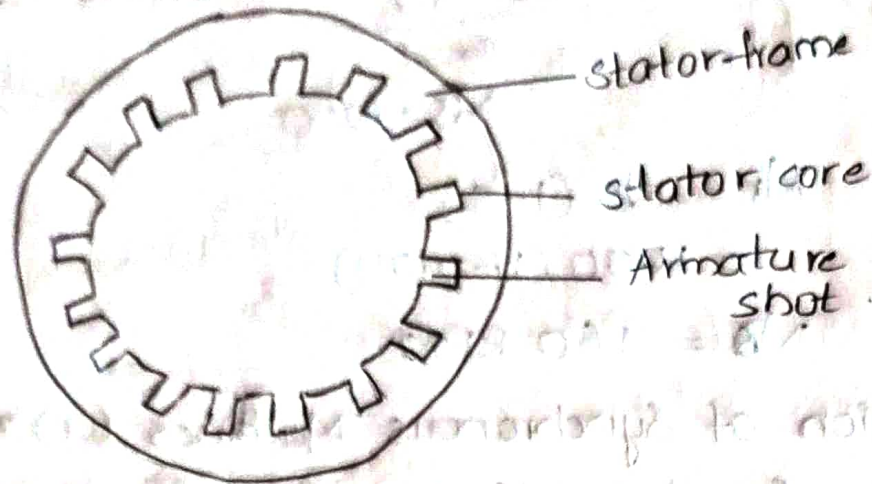
ii) Rotor [field winding]

iii) Stator



Stator:

Stator is the stationary part and cylindrical structure as shown in fig.



The stator carries Armature winding and this type of stator by using low rating material iron and cast iron used and high rating material 4% of Silicon steel is used which is Build up of the lamination 0.5mm which are insulated each other reduce the Eddy currents losses.

⇒ The stator core classified into 3 types

i) Open slot

Semi closed slot

closed slot

i) Open slot

The advantage of open slot is easy removal of replacement of detaching coils

ii) Semi closed slot:

It is generate the emf with less Re factor initial cost is low

closed slot: It is air gap is uniform.

Rotar:

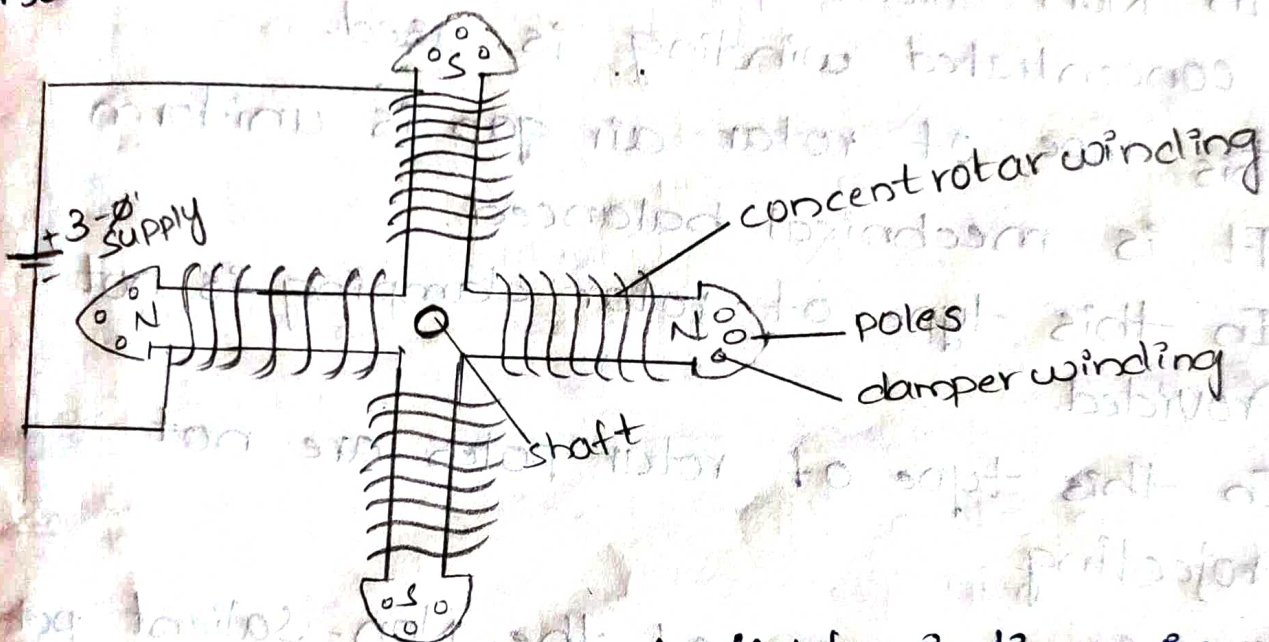
Rotar is a Rotating part and it carries fielding winding and produce the magnetic flux.

→ The Rotar is classified into two types.

- 1) Salient pole rotar (or) Non-cylindrical rotar
- 2) Non salient pole rotar (or) cylindrical rotar.

1) Salient pole rotar:

The Salient pole rotar as shown in fig.



In salient pole rotar of field winding is concentrated winding is used.

⇒ This type of rotar air gap is non uniform.

⇒ It is not mechanical balanced.

⇒ In this type of rotar damping is provided.

⇒ In this type of rotar poles are projecting.

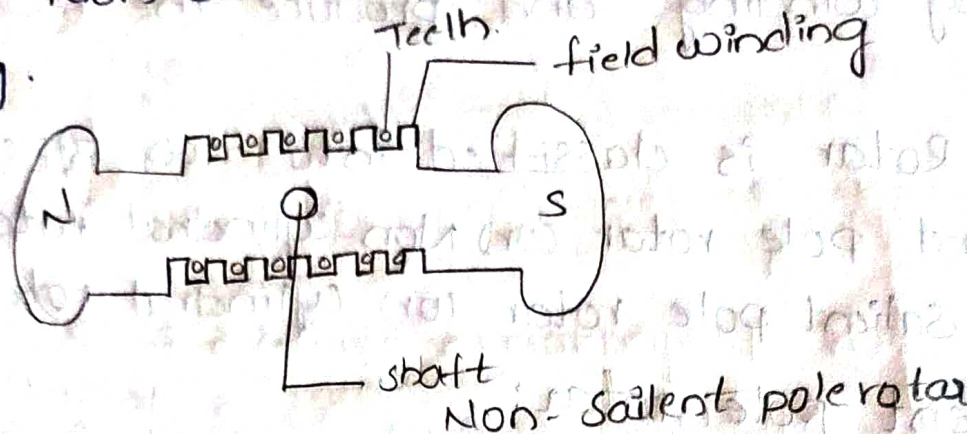
⇒ The speed range of the salient pole rotar is 50 - 1000 RPM.

⇒ In this type of rotar larger diameter and smaller axial length.

⇒ It is used in water turbines and I.E. engines.

Non salient pole rotor.

In Non salient pole rotor as shown in fig.



In Non salient pole rotor of field winding is concentrated winding is used.

⇒ This type of rotor air gap is uniform

⇒ It is mechanical balanced.

⇒ In this type of rotor damping is not provided.

⇒ In this type of rotor poles are not projecting

⇒ The Speed range of the Non-salient pole rotor is 1500-3000 RPM.

⇒ In this type of rotor smaller diameter and larger axial length.

Note:

Damper winding is used in preventing hunting in alternator to provide starting torque of electrical motors and shaft is used in transformation of power [They receiving dc power and deliver the AC power]

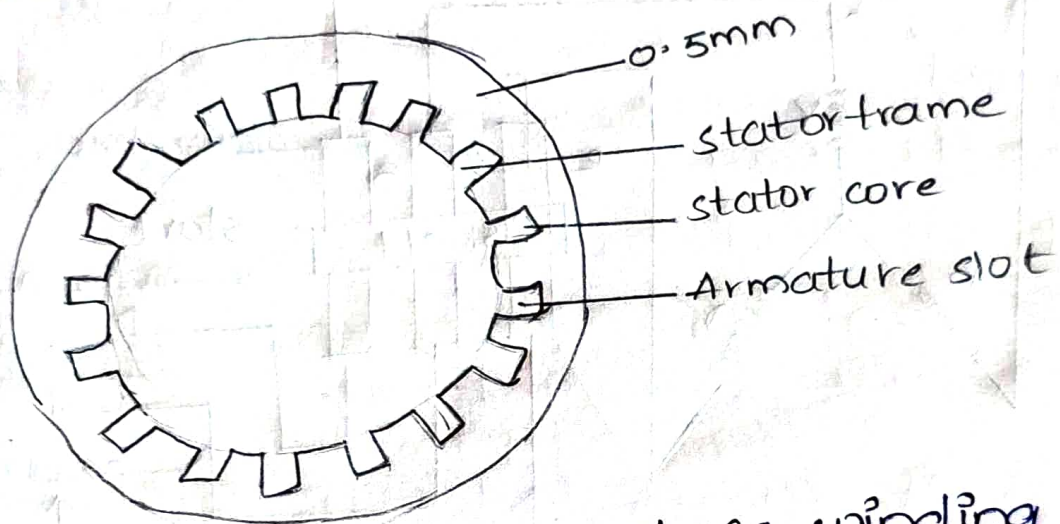
Construction of 3 phase Induction motor

The 3-phase Induction motor is self starting device and consists of two parts

- i) Stator
- ii) Rotor

Stator:

The stator is stationary part and cylindrical structure as shown in fig



The stator carries 3-phase winding & 3 phase supply is connected and this type of stator 4% of silicon steel is used

The stator is definite no of poles depends upon speed of the motor. If the no. of poles increases and speed of motor the increases.

The thickness of the lamination 0.5mm is used which are insulated by reducing Eddy current losses

Rotar

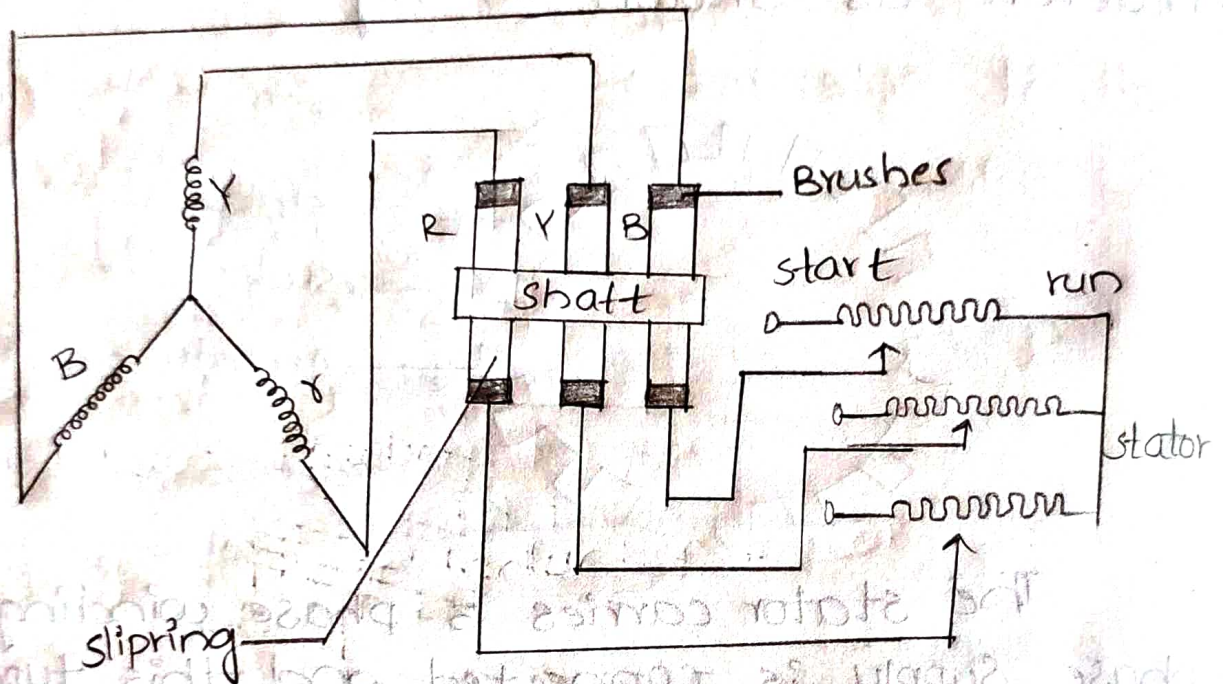
Rotar is a Rotating part and classified into two types

- i) slip ring rotar
- ii) Squirrel cage rotar

i) Slip ring rotar:

The slip ring rotar as shown in

fig



In this type of Rotar consisting of 3 phase supply similar to the stator.

⇒ In this type of rotar external resistance will be added.

⇒ It is complex construction and high cost

In this type of rotar maintenance cost is more

⇒ It is low efficiency and low power factor

⇒ In this type of rotar slip ring and brushes are present

It is high starting torque is applied.
In this type of rotor copper losses is high due to external resistance
In this type of rotor is used in lift and cranes.

Speed Advantages

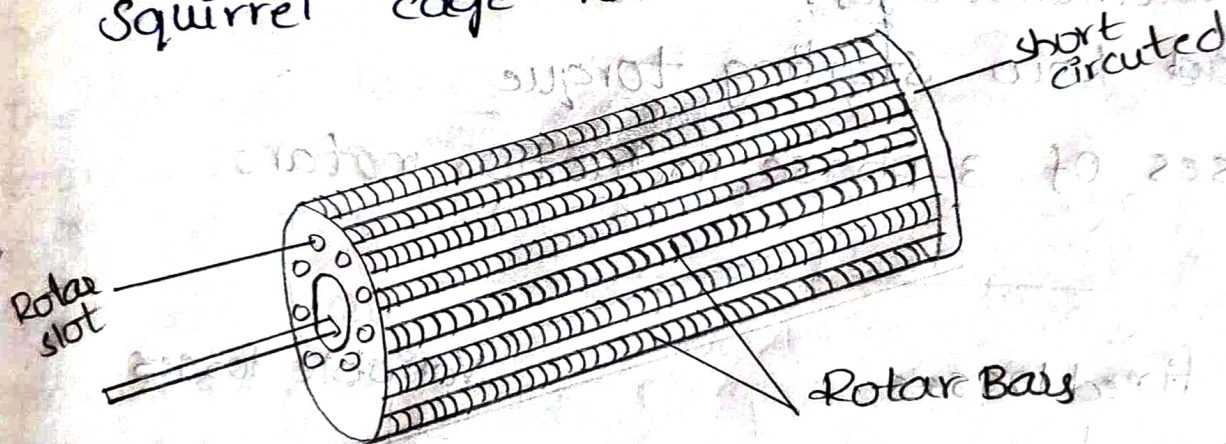
- Speed is constant
- Low starting current
- High starting torque

Disadvantages

- Low efficiency
- Low power factor.

Squirrel cage rotor:

Squirrel cage rotor as shown in fig.



In this type of rotor consisting of Bars which are short circuited with the help of end rings.

In this type of rotor external resistance can't be added.

It is simple construction is low cost.

In this type of rotor maintenance cost is high.

- ⇒ It is high efficiency and high power factor.
- ⇒ In this type of rotar slip ring and brushes are absent
- ⇒ It is low starting torque ~~can be~~ can't be obtained.
- ⇒ In this type of rotar copper losses is low due to external resistance
- ⇒ In this type of rotar is used in fans and fridges
- ⇒ speed control is not possible.

Advantages:

Low cost

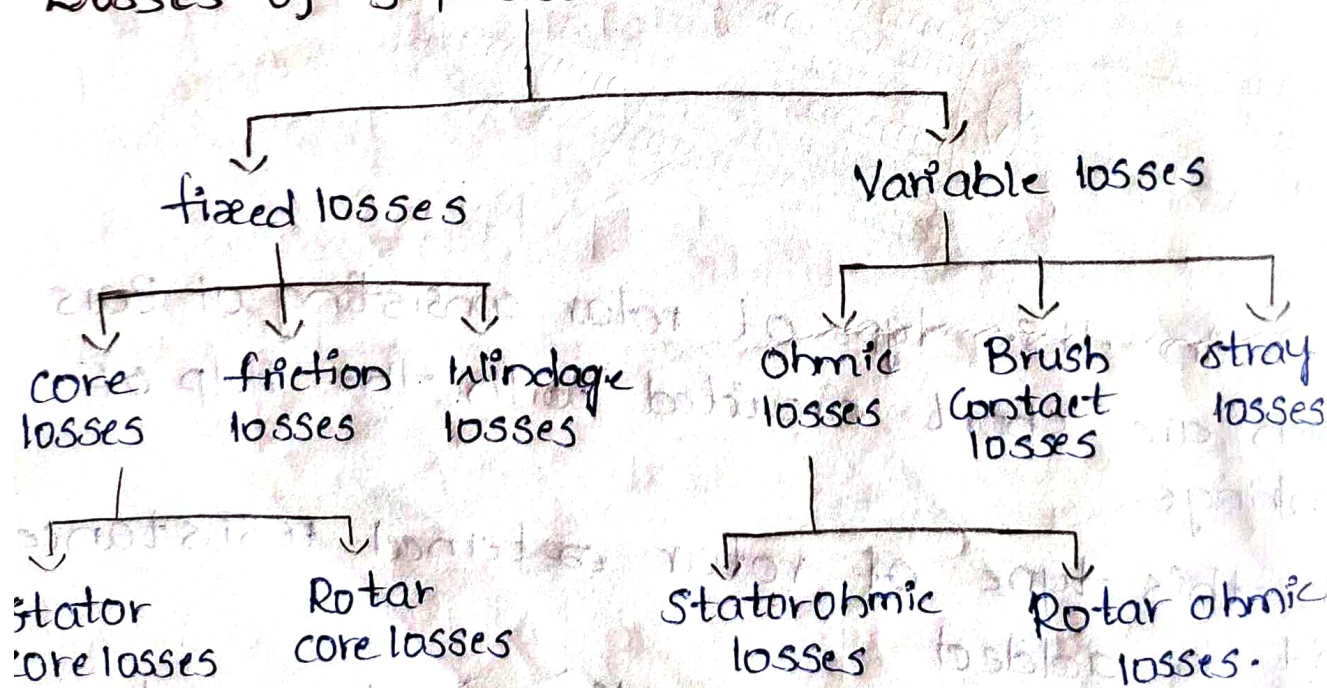
More efficiency

High overload capacity

Disadvantages:

low starting torque

Losses of 3-phase Induction motor:



- ⇒ The losses occurs whenever the motor remains constant for any speed and voltage is known as fixed losses.

→ The fixed losses occurs speed of the motor changes very slightly and compare to the speed of the induction motor. The frequency value will be decrease.

→ The core losses also divided into stator core losses and Rotor core losses and friction and windage losses are constant.

→ The no load test of three phase induction motor determine the fixed losses

fixed losses
Variable losses

→ The losses occurs whenever the motor load increases and the losses also increases is known as Variable losses.

→ The ohmic losses known as I^2R losses.

⇒ The ohmic losses occurs resistance of the stator and Rotor winding and generate the heat in increasing temperature.

⇒ Load test of 3-phase induction motor (or) back rotor test of 3-phase induction motor determine the variable losses.

Efficiency of 3-phase Induction motor:
It is defined as the ratio of mechanical energy output to electrical energy input is known as efficiency of 3-phase induction motor.

The mathematical expression of the 3-Phase induction motor is given as

$$\% \eta = \frac{\text{Mechanical energy output (Rotor)}}{\text{Electrical energy input (stator)}} \times 100$$

Let consider

P_{shatt} = rotor out power of induction motor

$P_{stator\ core}$ = stator core losses

$P_{rotor\ core}$ = rotor core losses

$P_{fric\ win}$ = friction losses + windage losses

$P_{I^2R\ stator}$ = stator ohmic losses

$P_{I^2R\ rotor}$ = Rotor ohmic losses

$P_{additional}$ = Brush contact losses + stray losses

$$P_c = P_{stator\ core} + P_{rotor\ core} + P_{fric\ win}$$

$$P_{I^2R} = P_{I^2R\ stator} + P_{I^2R\ rotor} + P_{addl}$$

$$\eta = \frac{P_{out}}{P_{in}}$$

$$P_{out} = \text{Input} - \text{Total losses}$$

$$P_{in} = \text{o/p} + \text{Total losses}$$

$$\text{Total losses} = P_c + P_{I^2R}$$

The efficiency of 3 phase induction motor

is given as

$$\% \eta = \frac{P_{shatt}}{P_{shatt} + (P_c + P_{I^2R})} \times 100$$

Q) 3-phase induction motor 6 poles 440 V and 50 Hz
the rotor input power 60 Kilowatts it is observed
that the rotor emf 90 complete cycle per minute
calculate the

- i) slip
- ii) Rotor Speed
- iii) Rotor copper losses per phase
- iv) Mechanical power develop
- v) Rotor resistance per phase with current $I_R = 60 \text{ Amp}$

Given data

3- ϕ induction motor

No. of poles = 6

voltage $V = 440 \text{ V}$

frequency = 50 Hz

Rotor Input power = 60 kW

Rotor emf = 90 cycle/minute

i) $S = ?$

ii) $N_r = ?$

iii) Rotor cu per phase = ?

iv) $P_m = ?$

v) $R_r = ?$

$$f_r = \frac{90}{60}$$

$$f_r = 1.5 \text{ Hz}$$

i) $f_r = S \times f$

$$S = \frac{f_r}{f}$$

$$= \frac{1.5}{50}$$

$$S = 0.03$$

ii) $N_r = N_s (1 - S) \text{ rpm}$

$$N_s = \frac{120 f}{p}$$

$$= \frac{120 (50)}{6}$$

$$N_s = 1000 \text{ rpm}$$

$$I_R \times 60 \text{ A} = 49$$

$$\begin{aligned} 49 \times 1000 &= 49000 \\ \frac{49000}{60} &= 816.67 \end{aligned}$$

$$f_r = 1.5 \text{ Hz}$$

$$N_r = 1000(1-0.03)$$

$$N_r = 970 \text{ rpm}$$

$$\begin{aligned} \text{iii) Rotor copper per phase} &= \frac{\text{Rotor I/p power} \times \text{slip}}{3} \\ &= \frac{60 \times 10^3 \times 0.03}{3} \end{aligned}$$

$$\text{Rotor copper losses per phase} = 600$$

iv) Mechanical power develop

$$P_m = P_{in}(1-s)$$

$$P_m = 60(1-0.03)$$

$$P_m = 58.2 \text{ KW}$$

v) Rotor copper losses per phase

$$I_r^2 R_r$$

$$600 = (60)^2 \times R_r$$

$$600 = 3600 \times R_r$$

$$R_r = \frac{600}{3600}$$

$$R_r = 0.16 \Omega$$

Q) 3- ϕ induction motor 50 Hz, 600V 9 voltage and 4 poles rotor input power 75 Kilowatt it is observed the rotor emf 120 complete cycle per min. calculate the

i) Slip

ii) Rotor Speed

iii) Rotor Copper losses

iv) Mechanical power develop

v) Rotor resistance with $I_r = 80 \text{ Amp}$

Given data

3- ϕ induction motor

No. of poles = 4

Voltage $V = 9$

frequency = 50 Hz

Rotor input power = 75 Kw

Rotor emf = 120 cycle / minute

i) $s = ?$

ii) $N_r = ?$

iii) Rotor cu per losses = ?

iv) $P_m = ?$

v) $R_r = ?$

$$f_r = \frac{120}{60}$$

$$f_r = 2 \text{ Hz}$$

i) $f_r = s \times f$

$$s = \frac{f_r}{f}$$

$$= \frac{2}{50}$$

$$s = 0.04$$

ii) $N_r = N_s (1 - s)$

$$N_s = \frac{120 f}{P}$$

$$= \frac{120(50)}{4}$$

$$= \frac{6000}{4}$$

$$N_s = 1500$$

$$N_r = N_s (1 - s)$$

$$= 1500 (1 - 0.04)$$

$$N_r = 1440 \text{ rpm}$$

$$\text{iii) Rotor copper losses} = \text{Rotor } I^2 R \times s$$

$$= 75 \times 10^3 \times 0.04$$

$$\boxed{\text{Rotor copper losses} = 3000 \text{ W}}$$

iv) Mechanical power

$$P_m = P_{in} (1-s)$$

$$= 75 (1-0.04)$$

$$\boxed{P_m = 72 \text{ kW}}$$

$$\text{v) Rotor copper losses} = I_R^2 \times R_r$$

$$= (80)^2 \times R_r$$

$$3000 = 6400 \times R_r$$

$$R_r = \frac{3000}{6400}$$

$$\boxed{R_r = 0.46 \Omega}$$

Q) 3- ϕ induction motor 50-HZ the line voltage 500 V and 6 poles gives an ~~rotor~~ output power 37.3 kW at 955 rpm the power factor 0.86 frictional and windage losses 1.492 kW the stator losses approximately 1.5 kW. determine the line current, efficiency and rotor copper losses this motor.

Given data

3- ϕ induction motor

frequency $f = 50 \text{ Hz}$

Line voltage $V_L = 500 \text{ V}$

No. of poles = 6

Rotor out put = 37.3 kW

$N_r = 955 \text{ rpm}$

$\cos \phi = 0.86$

$$s = \frac{N_s - N_r}{N_s}$$

$$N_s = \frac{120f}{P}$$

$$\boxed{N_s = 1000}$$

$$s = \frac{1000 - 955}{1000}$$

$$\boxed{s = 0.045}$$

Rotor copper

Mech

Rotor

Rotor

Rotor

Rotor

Rotor

Rotor

Rotor

$$\% \eta = \frac{P_m}{P_{in}} \times 100$$

stator in

stator in

$$\% \eta = \frac{P_m}{P_{in}} \times 100$$

$$i) \text{ Rotor copper losses} = \text{Rotor } I^2 R \times \text{slip}$$

$$= 75 \times 10^3 \times 0.04$$

$$\boxed{\text{Rotor copper losses} = 3000 \text{ W}}$$

ii) Mechanical power

$$P_M = P_{in} (1 - s)$$

$$= 75 (1 - 0.04)$$

$$\boxed{P_M = 72 \text{ kW}}$$

$$v) \text{ Rotor copper losses} = I_R^2 \times R_r$$

$$= (80)^2 \times R_r$$

$$3000 = 6400 \times R_r$$

$$R_r = \frac{3000}{6400}$$

$$\boxed{R_r = 0.46 \Omega}$$

$$\boxed{511.8 \text{ A} = 11}$$

Q) 3- ϕ induction motor 50-HZ the line voltage 500 V and 6 poles gives an ~~ap~~ ^{rotor} output power 37.3 kW at 955 rpm the power factor 0.86 frictional and windage losses 1.492 kW the stator losses approximately 1.5 kW. determine the line current, efficiency and rotor copper losses this motor.

Given data

3- ϕ induction motor

frequency $f = 50 \text{ Hz}$

Line voltage $V_L = 500 \text{ V}$

No. of poles = 6

Rotor output = 37.3 kW

$N_r = 955 \text{ rpm}$

$\cos \phi = 0.86$

frictional and windage losses = 1.492 kW

Stator losses = 1.5 kW

$$s = \frac{N_s - N_r}{N_s}$$

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{6}$$

$$N_s = 1000 \text{ rpm}$$

$$s = \frac{1000 - 955}{1000}$$

$$s = 0.04$$

Rotar copper losses = Rotar I/p power $\times$ slip

Mechanical power

$$\frac{\text{Rotar output power}}{\text{Rotar input power}} = \frac{N_r}{N_s}$$

$$\text{Rotar I/p} = \frac{\text{Rotar o/p} \times N_s}{N_r}$$

$$= \frac{37.3 \times 10^3 \times 1000}{955}$$

$$\text{Rotar I/p} = 39057.5 \text{ W}$$

$$\text{Rotar I/p} = 39.05 \text{ kW}$$

$$\text{Rotar copper losses} = 39.05 \times 10^3 \times 0.04$$

$$= 1562 \text{ W}$$

$$\text{Rotar copper losses} = 1.562 \text{ kW}$$

$$\% \eta = \frac{\text{Mechanical energy output power}}{\text{electrical energy input power}} \times 100$$

$$\leftarrow \text{stator input power} = \text{Rotar input power} + \text{stator losses}$$

$$= 39.05 + 1.5$$

$$\text{stator input power} = 40.55 \text{ kW}$$

$$\% \eta = \frac{37.3}{40.55} \times 100$$

$$\% \eta = 91.98\%$$

Q) $I_L = ?$

$$\text{Stator input power} = \sqrt{3} V_L I_L \cos \phi$$

$$40.55 \times 10^3 = \sqrt{3} \times 500 \times I_L \times 0.86$$

$$I_L = \frac{40.55 \times 10^3}{\sqrt{3} \times 500 \times 0.86}$$

$$I_L = 54.44 \text{ A}$$

Q) 3- ϕ induction motor 50 Hz, 600 V line voltage and 6 poles the rotor output power 47.5 Kw at 800 RPM. The power factor 0.45 the frictional and windage losses 1.7 Kw and stationary losses 1.75 Kw approximately. Determine the line current efficiency rotor copper losses of the motor.

Given data

3- ϕ induction motor

frequency = 50 Hz

line voltage = 600 V

No. of poles = 6

rotor output power = 47.5 Kw

$N_r = 800 \text{ RPM}$

$\cos \phi = 0.45$

frictional and windage losses = 1.7 Kw

$$s = \frac{N_s - N_r}{N_s}$$

$$N_s = \frac{120 f}{P} = \frac{120 \times 50}{6}$$

$$N_s = 1000 \text{ rpm}$$

$$S = \frac{1000 - 800}{1000} = \frac{200}{1000}$$

$$S = 0.2$$

rotor copper losses = Rotor I/p power $\times$ slip
Mechanical power

$$\frac{\text{Rotor o/p power}}{\text{Rotor I/p power}} = \frac{N_r}{N_s}$$

$$\text{Rotor I/p} = \frac{\text{rotor o/p} \times N_s}{N_r}$$

$$= \frac{47.5 \times 1000 \times 10^3}{800}$$

$$= 59375 \text{ W}$$

$$= 59.37 \text{ kW}$$

$$\text{Rotor copper losses} = 59.37 \times 10^3 \times 0.2$$

$$= 11874 \text{ W}$$

$$= 11.874 \text{ kW}$$

$$\therefore \eta = \frac{\text{mechanical energy o/p power}}{\text{electrical energy I/p power}} \times 100$$

$$\text{stator I/p power} = \text{Rotor I/p power} + \text{stator losses}$$

$$= 59.37 + 1.7$$

$$= 61.07$$

$$\therefore \eta = \frac{47.5}{61.07} \times 100$$

$$= 77.7\%$$

$$I_L = ?$$

$$\text{stator I/p power} = \sqrt{3} V_L I_L \cos \phi$$

$$61.07 \times 10^3 = \sqrt{3} \times 600 \times I_L \times 0.45$$

$$I_L = \frac{61.07 \times 10^3}{\sqrt{3} \times 600 \times 0.45}$$

$$I_L = 120.70 \text{ A}$$

Construction of Single phase Induction Motor

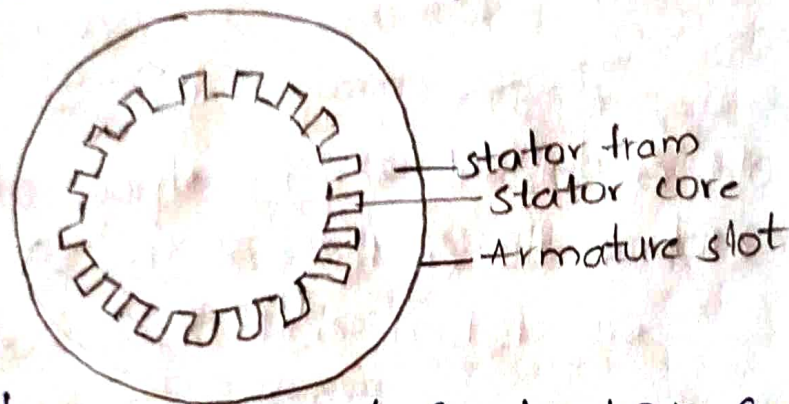
⇒ The Single phase induction motor is not self starting device and it consisting of two parts

i) Stator

ii) Rotor

i) Stator

Stator is a stationary device and cylindrical structure as shown in fig.



The stator carries 1- ϕ Supply two connected and which are insulated by ~~varnished~~ lamination of 4% of Silicon steel is used in this material and thickness of the lamination from 0.036 - 0.06 cm and reduce the Eddy current losses.

Rotor:

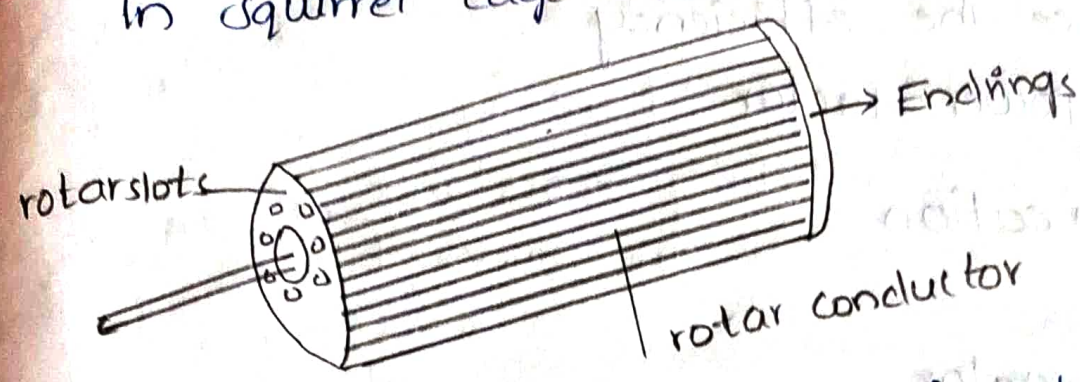
Rotor is a rotating part and in classification rotor are

i) Squirrel cage rotor

ii) Slip ring rotor

Squirrel cage rotor

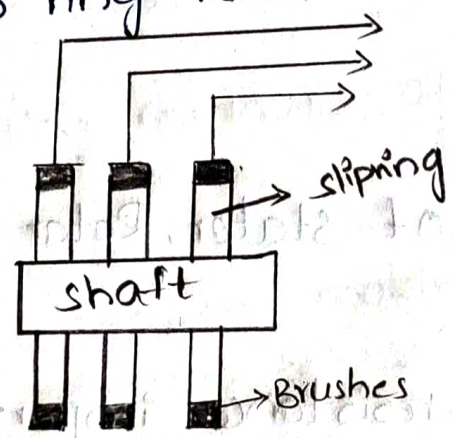
In squirrel cage rotor as shown in fig



In type of rotor consisting of rotar slots to the lamination of the silicon steel is used and rotar slots connected in series which carries the rotar bars and rotar conductors connected to end rings it reduce the copper losses is low due to low power-factor.

ii) Slip ring rotor

In slip ring rotor as shown in fig



In this type of rotor consisting of sliprings are connected rotor and shaft.

⇒ The slip rings connected to the shaft in Parallel connection and reduce the copper losses and high power-factor

Advantages:

- i) Most economical
- ii) easy to repair

Disadvantages

- i) Reduce the efficiency
- ii) Low power factor

Application

- i) fans
- ii) Refrigerator
- iii) Vacuum cleaners
- iv) Printing machines
- v) Washing machines
- vi) Mixers, Grinders

factors of single phase Induction Motor

The classification of factors of single phase induction motor are

- i) Rotor resistance
- ii) Air gap length
- iii) physical structure of stator, Rotor
- iv) Rotor resistance

The rotor resistance important parameters of performance of $1-\phi$ induction motor which are determine with help of area of cross section of the material.

If the rotor resistance increases and the torque will be maximum.

If the rotor resistance increases decrease the starting current and efficiency and percentage

of voltage regulation is increases.

Air gap length:

Reduction of the air gap length it causes more losses and reduce the overload capacity.

The air gap length completely depends upon power factor and no. of poles.

Physical structure of stator, Rotor.

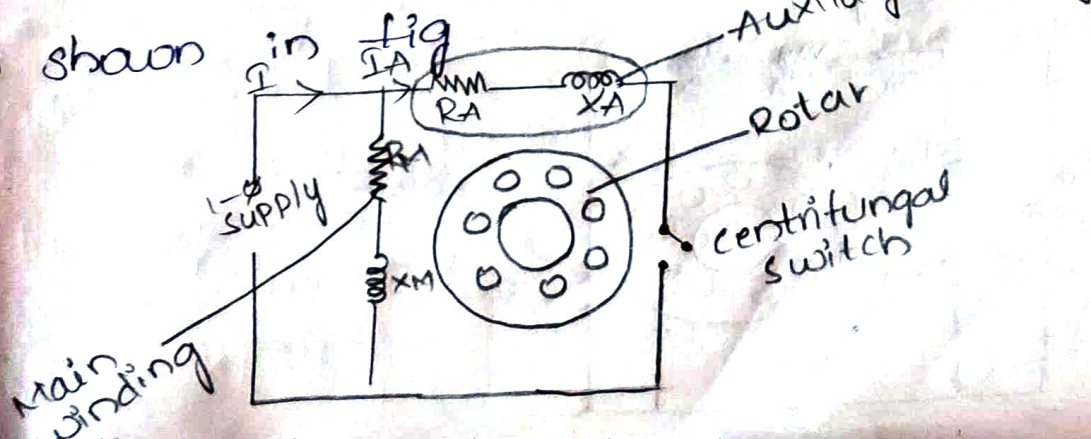
These parameters Very important role of working of single phase induction motor the maximum torque will be produced reduce the intergal slot winding and increase the air gap length.

Starting methods / classification of 1- ϕ induction motor.

The starting methods of 1- ϕ induction motor.

- i) Split phase Induction motor
- ii) capacitor start induction motor
- iii) capacitor run Induction motor
- iv) Two value capacitor Induction motor
- v) Split phase induction motor;

In type of motor in the circuit as shown in fig



In type of induction motor connected in single phase supply and flow of current divided into auxiliary and main winding current.

⇒ In auxiliary winding resistance, reactance and centrifugal switch are connected in series and parallel with main winding.

⇒ In this type of motor has low efficiency, low power-factor and slow starting torque

Start characteristic of Split phase induction motor:-

i) The starting torque of the motor 100-250%.

ii) Break down torque of the motor upto 300%.

iii) The efficiency of motor 55-65%.

iv) The power-factor of the induction motor 0.5-0.65.

v) The power rating induction motor range from $\frac{1}{2}$ HP - 1 HP.

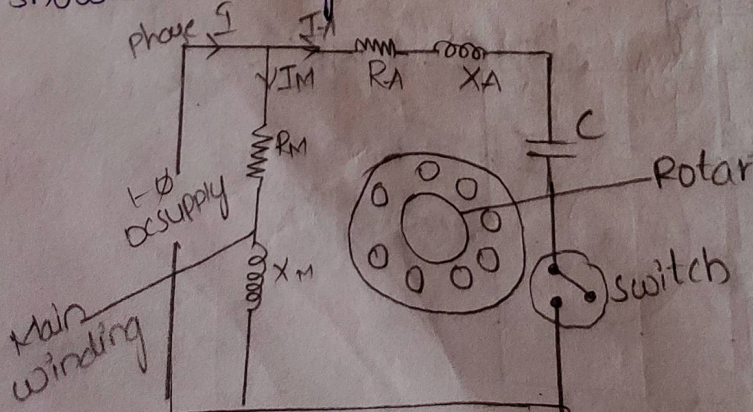
Applications:

i) Washing machines

ii) Mixers, Grinders

Capacitor start induction motor.

In this type of motor in the circuit as shown in fig.



In this type resistance, reactance in series with capacitor with main winding.

In this type of and low power

characteristic of motor.

i) The starting 250-400%.

ii) Breakdown t

iii) The efficiency

iv) The power-fac 0.5-0.65%.

v) The power from $\frac{1}{8}$ HP

Applications

i) Motor pur

ii) Air comp

Capacitor

as shown

In type of induction motor connected in single phase supply and flow of current divided into auxiliary and main winding current.

⇒ In auxiliary winding resistance, reactance and centrifugal switch are connected in series and parallel with main winding.

⇒ In this type of motor has low efficiency, low power factor and slow starting torque.

~~Start~~
characteristic of Split phase induction motor:-

i) The starting torque of the motor 100-250%.

ii) Break down torque of the motor upto 300%.

iii) The efficiency of motor 55-65%.

iv) The power factor of the induction motor 0.5-0.65.

v) The power rating induction motor range from $\frac{1}{2}$ HP - 1 HP.

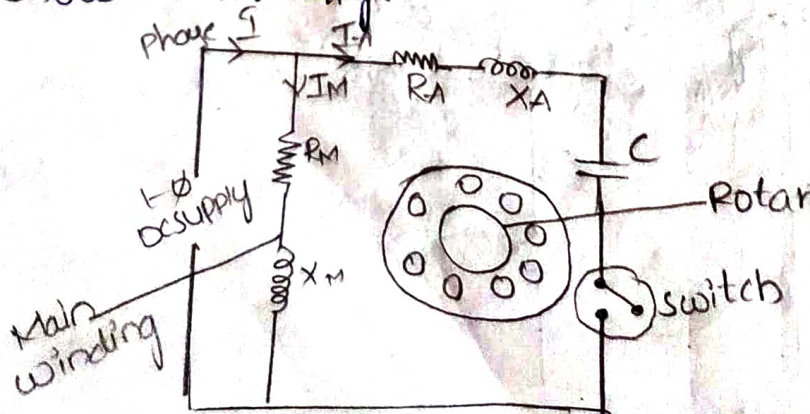
Applications:

i) Washing machines

ii) Mixers, grinders

Capacitor start induction motor.

In this type of motor in the circuit as shown in fig.



In this type of motor auxiliary winding resistance, reactance and capacitor connected in series with centrifugal switch and parallel with main winding.

In this type of motor also low efficiency and low power-factor.
characteristic of capacitor start induction motor.

1) The starting torque of the induction motor 250-400%.

2) Breakdown torque of the motor upto 350%.

3) The efficiency of motor 55-65%.

4) The power-factor of the induction motor 0.5-0.65%.

5) The power rating of induction motor range from 1/8 HP - 1 HP.

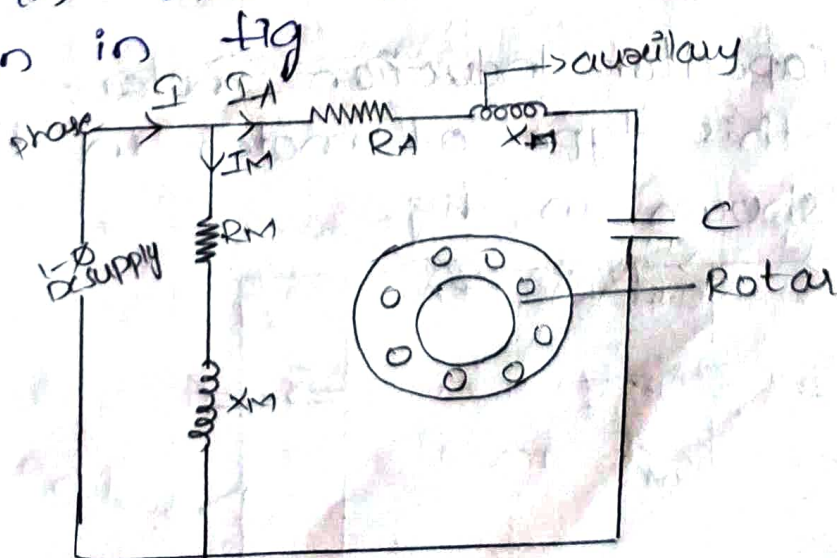
Applications:

1) Motor pump

2) Air compression

Capacitor Run induction motor:

In this type of motor in the circuit as shown in fig



In this type of motor in auxiliary resistance, reactance and capacitance are connected in series in the circuit no centrifugal switch is required and parallel with main winding.

In this type of motor high starting torque, high power-factor and high efficiency.

Characteristic Capacitor run Induction motor:

i) The starting torque of the induction motor

50-100%

ii) Break down torque of the motor upto

250

iii) Efficiency of the motor 60-70

iv) The power factor of the induction motor

0.75-0.9

v) The power rating induction motor range from 1/8 HP - 1 HP

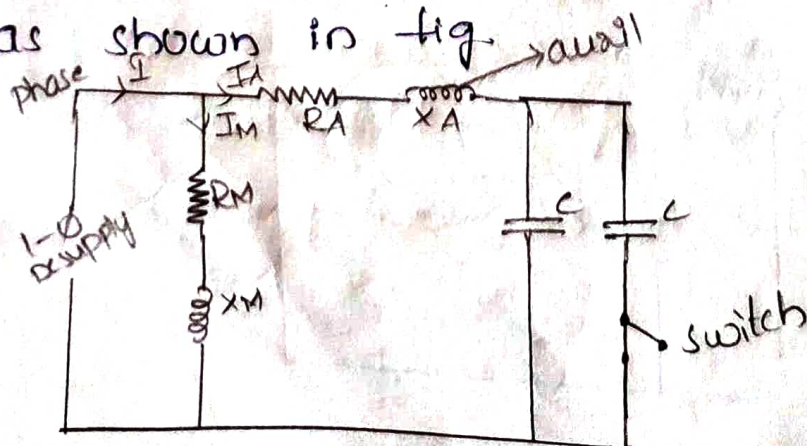
Applications:

i) fans

ii) voltage regulator.

Two value Capacitor Induction motor:

In this type of motor in the circuit as shown in fig.



In this type of motor in auxiliary winding resistance, reactance and capacitance CR is series connection

CR = Permanent circuit are parallel with CS where CS = short-time capacitor in the circuit and parallel with main winding.

In this type of motor high starting torque, high power-factor and high efficiency characteristic of two value capacitor induction motor are.

↓ The starting torque of the induction motor 200-300%

ii) Breakdown torque of the motor upto 250%

iii) The power efficiency of the induction motor 60-70%

iv) The power factor of the induction motor 0.75-0.9

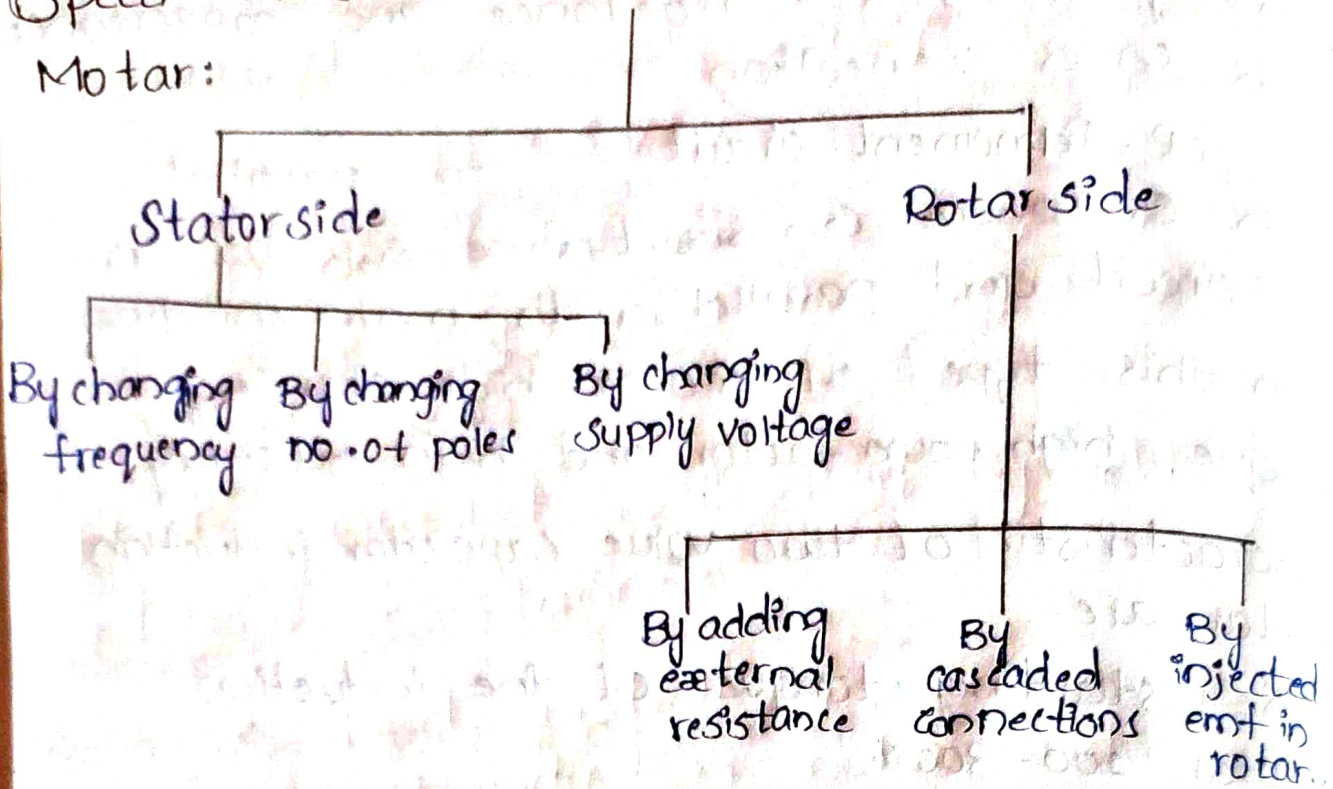
↓ The power rating induction motor range from 1/8 HP - 1 HP

Applications:

i) Refrigerator

ii) Air condition compression.

Speed controlled Methods of 3-phase Induction Motor:



The Various factors of Speed controlled motor are 3- ϕ induction motor

⇒ Speed controlled By changing frequency:

- i) In the Speed controlled method the Speed is directly proportional to Supply frequency. If the Supply frequency changes the speed of the motor changes simultaneously.
- ii) In Speed Control method the Supply voltage is constant the frequency value decreases and flux is increases.
- iii) If the flux is increases reduce the iron losses and efficiency.
- iv) If the frequency increases the flux value will be decreases and reduce the maximum torque

Speed controlled method by changing no. of poles:

It is suitable for only squirrel cage induction motor because squirrel cage rotor is automatically developed no. of poles equal to poles of the stator winding.

Suppose motor has two winding for 6 poles and 4 poles and 50 Hz supply the synchronous speed will be 1000 - 1500 RPM. respectively and for slip value 5% the rotor speed will be 950 and 1425 RPM respectively.

i) By changing no. of poles increases and the rotor and synchronous speed will be decreases.

ii) In the Speed controlled method it is more costly.

Speed controlled method by changing supply voltage:

i) In this method of speed controlled is constant frequency and varied the supply voltage

ii) The torque will be produced by induction motor which is directly proportional to square of rotor induced emf

$$T \propto E^2$$

iii) This rotor induced emf is directly proportional to supply voltage

$$E_2 \propto V$$

iii) The torque is directly proportional to square of the supply voltage is given as

$$T \propto S V^2$$

from based on the above eq supply voltage decreases and the torque value also decreases.

Speed controlled method of by adding external resistance:-

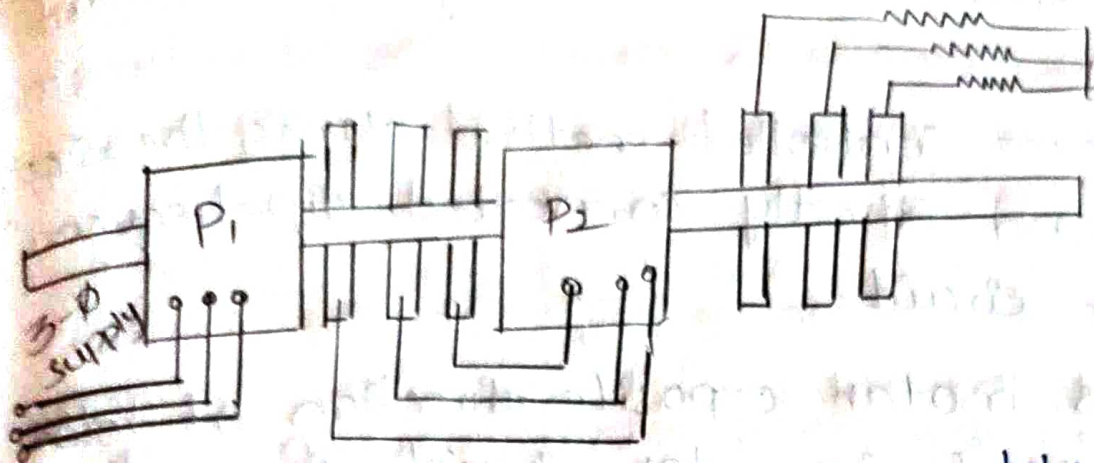
i) In the Speed controlled method suitable for slip ring induction motor and the Speed controlled method similar to Speed control of Dc shunt motor.

ii) In the Speed control method speed is valued by adding external resistance in rotor circuit the starting torque will be increases it causes more losses.

iii) In the Speed control method - frequency and Supply voltage is constant and the Speed of the motor continuously decreases.

Speed controlled method by cascaded connection

i) If the Speed controlled method basically arrangements of both motor connected signal shaft then they are motor run same side of the rotor



for P₁ motor can be run separately from the synchronous speed will be

$$N_s = \frac{120f}{P_1}$$

for P₂ motor can be run separately from the synchronous speed will be

$$N_s = \frac{120f}{P_2}$$

Basically consisting of both motors connected in cascaded connection and the synchronous speed is

$$N_s = \frac{120f}{P_1 + P_2}$$

It is consisting of both motors connected in different mode based on the no. of poles the synchronous speed is given as

$$N_s = \frac{120f}{P_1 - P_2} \quad \text{or} \quad N_s = \frac{120f}{P_2 - P_1}$$

In the speed control method torque and speed of the shaft will be increases and the frequency and the synchronous speed decreases.

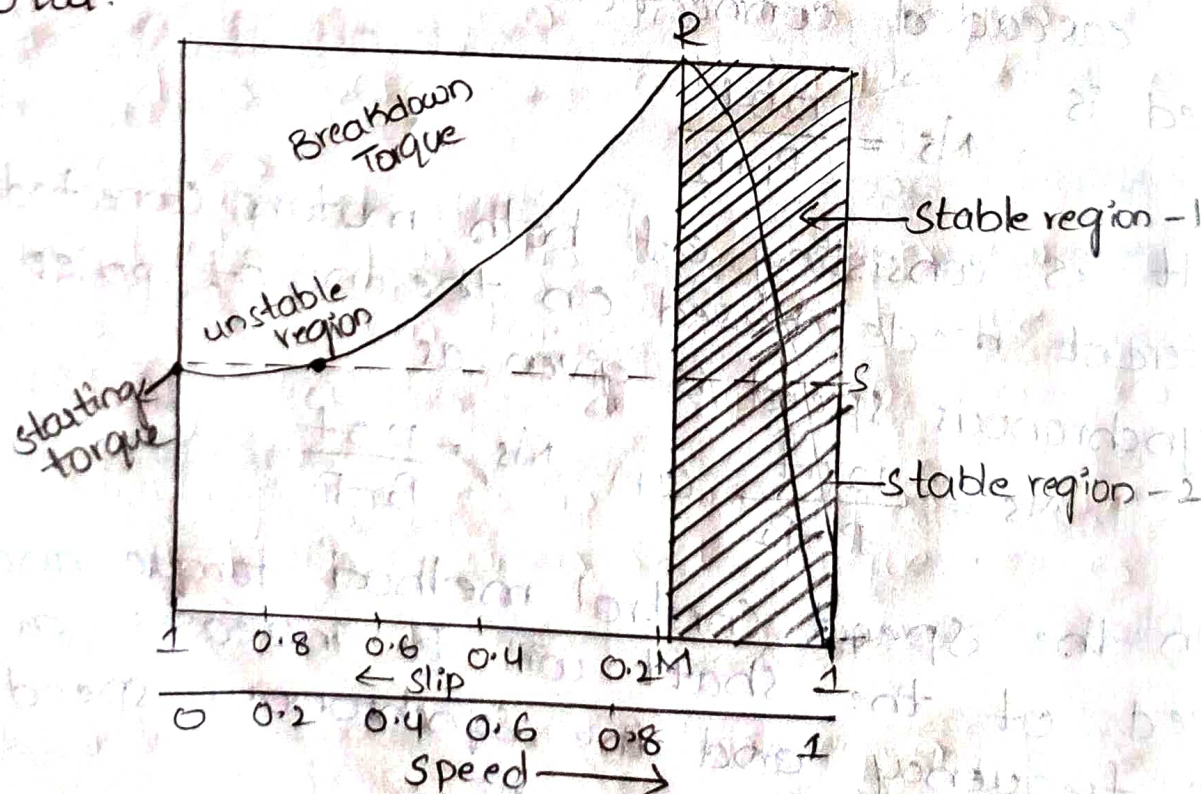
Speed Controlled method by injected emf in rotor.

In Speed Controlled method varies the supply voltage and directly connected slip frequency in rotor circuit.

In this motor opposite direction of rotor induced emf in rotor circuit the voltage resistance will be increases and the speed of the motor decreases.

In injected emf of rotor connected in motor the total resistance decreases and speed of the motor increases.

~~my~~ In this method speed control is possible
Torque-slip characteristic of 3-phase induction motor.



Torque Slip characteristic of 3- ϕ induction motor is given as

$$T \propto \frac{SE_2^2 R_2}{(R_2)^2 + (SX_2)^2}$$

$$T = \frac{KSE_2^2 R_2}{(R_2)^2 + (SX_2)^2}$$

Where

K = Rotar constant

S = Slip

E_2 = Rotar induced emf

R_2 = Rotar resistance

X_2 = Rotar reactance

For case 1: No load condition

In no load condition of 3- ϕ induction motor is rotar speed will be equal to zero

$[N_r = 0]$ Based on the above condition the slip is given as

$$S = \frac{N_s - N_r}{N_s}$$

$$S = \frac{N_s - 0}{N_s}$$

$$\boxed{S = 1}$$

When $S = 1$ torque reaches starting point is known as starting torque.

For case 2: Increasing load condition.

In Increasing load condition of 3- ϕ induction motor is torque is inversely proportional to slip

$$\boxed{T \propto \frac{1}{S}}$$

In this condition if the load is increases torque and speed will be decreases and slip is increases.

for case 3: Maximum torque condition

The maximum torque condition of 3- ϕ induction motor is torque is directly proportional to slip $T \propto s$

Based on the above eqⁿ if the increasing load and the rotor and slip also increases. the torque reaches max torque is known as ~~breakdown~~ ^{torque} for case 4: Different modes of 3- ϕ induction motor

3- ϕ induction motor

consisting of four different modes are follows

- i) PQ - Unstable region
- ii) ST - Stable region
- iii) OP - Starting Torque
- iv) MR - Maximum torque

i) PQ - Unstable region: In this region if the load is increases and torque value will be decreases.

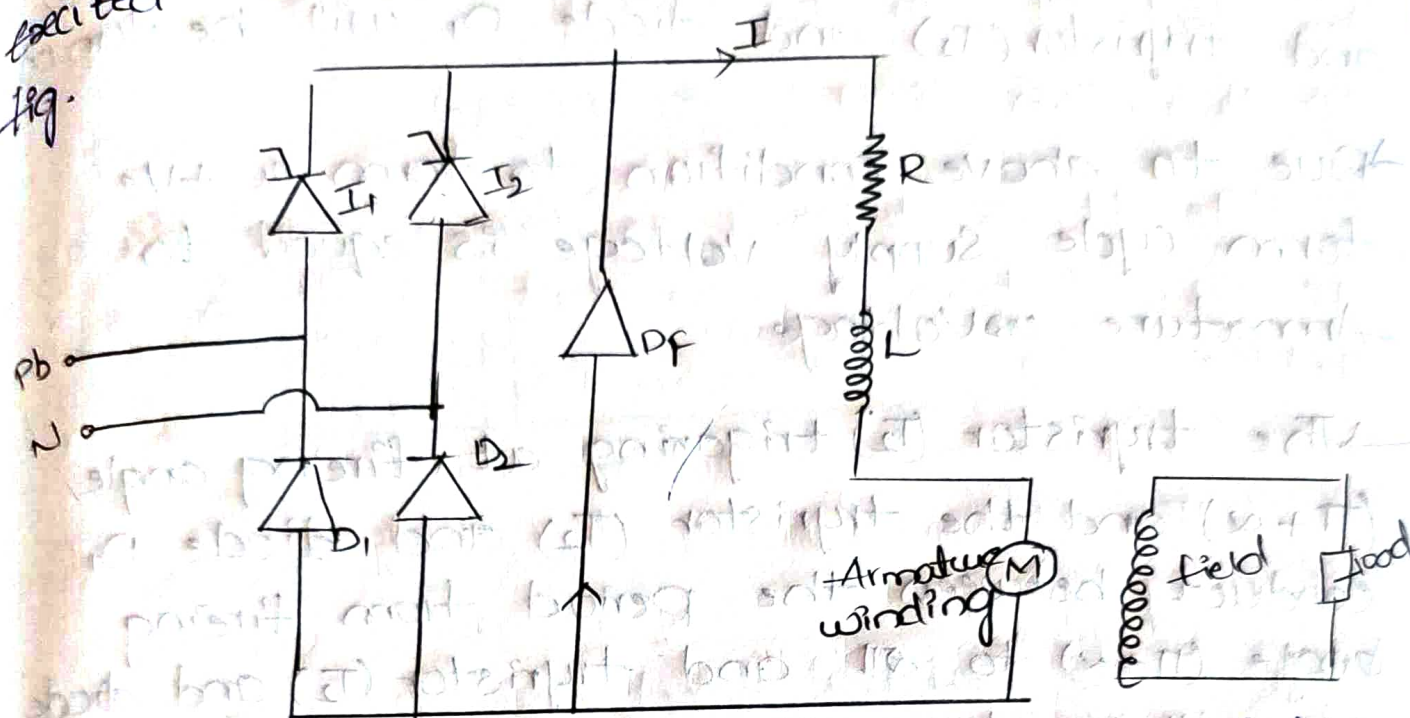
ii) ST - Stable region: In this region torque reaches maximum torque then torque continuously decreases and slip also decreases

iii) OP - Starting Torque: In this region whenever the applied load when $s=1$ torque reaches the starting point

iv) MR - Maximum Torque: In this region if the load is increases torque and slip also increases

Speed control method of Separately excited 1-phase D.C motor:

Fig.



In Speed control method of Separately excited single phase semi converter are controlled by Armature Voltage an Varying firing angle (α) and depending upon circuit parameters and operation

→ In the Speed control method the armature current can be continuous or discontinuous mode.

→ The effect of Armature current in discontinuous mode are given as

- i) Poor regulation
- ii) The power factor of the armature current will be increased

Hence by using 1- ϕ dc motor the continuous armature current can be operated

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→ The Thyristor (T_1) triggering at firing angle α and the thyristor (T_1) and diode D_2 conduct between the period from firing α to π and the thyristor (T_1) diode D_2 will be turn on and thyristor (T_2) and diode D_1 will be turn on

→ Due to above condition to form a +ve form cycle supply voltage is equal to Armature voltage

→ The thyristor (T_2) triggering at firing angle $(\pi + \alpha)$ and the thyristor (T_2) and diode D_1 conduct between the period from firing angle $(\pi + \alpha)$ to 2π and thyristor (T_2) and diode (D_1) will be turn on and thyristor (T_1) and diode (D_2) will be turn on

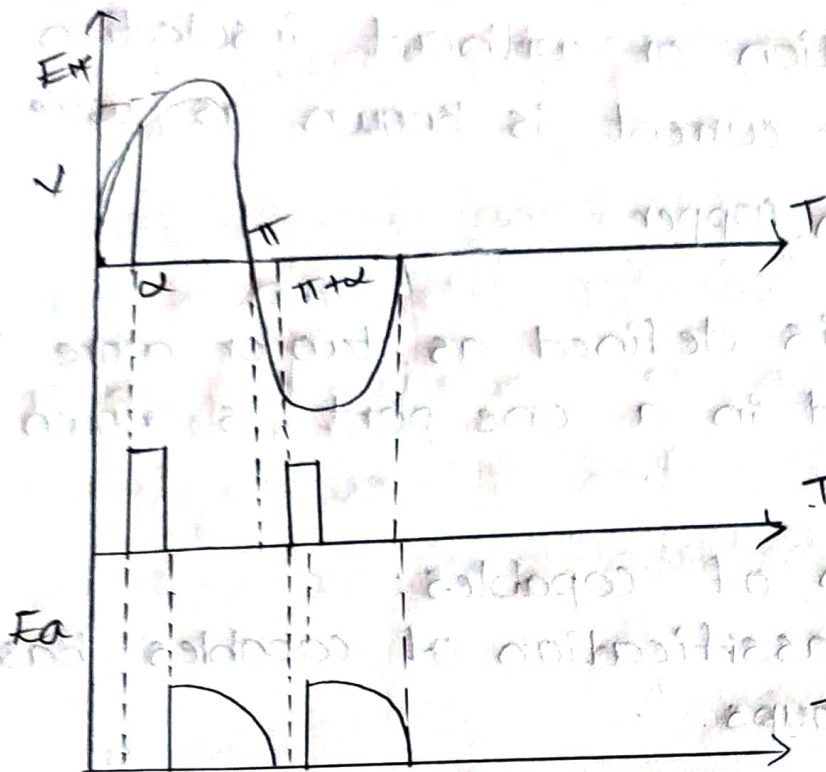
Due above condition to form a -ve of cycle supply voltage reverse to +ve of cycle.

→ The purpose of freewheeling Diode is used in circuit control the armature voltage in -ve of cycle and freewheeling diode and passing to the reverse direction of Armature Current and the armature voltage will be Zero.

→ The Average armature voltage is given as

$$E_a = \frac{E_m (1 + \cos \alpha)}{\pi}$$

The separately excited DC motor in supply voltage sinusoidal wave form as shown in fig



Torque Speed characteristic Separately excited d.c motor in this method the torque is directly proportional to torque increase speed also increases.

